thickness of the end fitting. Based on an x-ray inspection, the end fitting was threaded onto the end of the output rod such that the end of the rod was just aft of the inspection hole. Upon further inspection, the end of the rod was visible by looking through the inspection hole with an optical microscope.

Next, force required to move the cable through the conduit was measured. First, the conduit at the input end was held fixed, and the output end fitting was pulled with a force gauge. The output rod displaced approximately 0.30 inch with a force of 23 pounds. Next the conduit was held fixed while the control tube was pulled, and a displacement of approximately 0.33 inch was obtained with a force of 25 pounds.

After measuring the force to move the cable as described above, the cable was cut at the input end adjacent to the control tube to facilitate its removal from the conduit. Initial movement of the cable within the conduit occurred again at 23 pounds, but quickly increased to 43 pounds, approaching the capacity of the force gauge. The conduit was straightened by hand, and during the straightening process, wires of the conduit fractured. After straightening, approximately 2 feet of cable were removed with a force of less than 43 pounds. The cable was removed the rest of the way without measuring force.

On the cable, black charred material was observed 13 inches to 21 inches from the output end fitting attachment hole, and the conduit liner was melted and adhered to the cable at 33 inches to 38 inches from the output end fitting attachment hole consistent with heat damage to the liner. Additional cuts were made to inspect the output end of the assembly components between the conduit and the swivel tube. Results of the inspection showed further evidence of heat damage. Debris consistent with the conduit liner was observed near the output end of the conduit and appeared to have been displaced during removal of the cable. The debris was cleaned with acetone and air, and additional debris was removed with tweezers and cotton-tipped swabs. No obstructions were observed once the liner debris was removed.

2. Mixture Control Assembly

Closer views of the input and output ends of the mixture control assembly are shown in figure 8. As with the throttle control assembly, the positions of the mixture control assembly had been documented on-scene with a line of white paint on the release rod, control tube, and lock tube on the input end visible in the upper image in figure 8 and on the swivel tube and output rod on the output end visible in the lower image in figure 8. Remnants of an aluminum alloy housing was present at the aft end of the lock tube.

The mixture control position was measured at the output end of the assembly and compared to full rich and idle cut-off position descriptions provided by a Mooney representative. As installed on the airplane and measured from the clamp attach point to the center of the end fitting attachment hole (see the labeled bracket in figure 8), the full rich mixture position measures 7.70 inches, and the idle cut-off position measures